Smart Library Book Borrowing and Fine Tracking System for a Rwandan Community Library

Phase 0: Research and Proposal

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Background and Problem

- Community libraries in Rwanda use paper cards to track books.
- Books get lost because there is no clear record of who has them.
- Late fines are calculated by hand, causing mistakes and arguments.
- Members cannot know when a borrowed book will be available.

Motivation

- Reading helps students improve their education.
- Small libraries need a simple, digital system they can afford.
- I chose this project because paper systems waste time and lose data.

Objectives

- **Main:** Build an Oracle database to track books, borrowing, and fines.
- **Specific 1:** Block members who have 3 overdue books.
- **Specific 2:** Automatically calculate late fines in RWF per day.
- **Specific 3:** Manage a waiting queue when all copies of a book are out.
- **Specific 4:** Build a dashboard to show most borrowed books.

Research

- **Existing system (Koha):** It is a good library system, but it is too big and complex for a small Rwandan community library.

- **Gap:** Current systems do not easily handle local waiting lists or simple RWF fines for small libraries.

Data Source

- **Dataset:** Book-Crossing Dataset (Kaggle).
- **Link:** kaggle.com/datasets/ruchi798/bookcrossing-dataset
- **Size:** Over 278,000 real books, titles, authors, and years.
- **Reference Data:** I will add Rwandan member names, locations, and fine rates (100 RWF/day) because the dataset does not have them.

Proposed Solution & Innovation

- **Solution:** A centralized Oracle database to record borrowing, returns, and fines.
- **Innovation:** A Power BI dashboard connected to the database.
- **Dashboard shows:** Most borrowed books, members with unpaid fines, and waiting lists.

Planned Database Design

- MEMBER: Library members.
- BOOK: Book details.
- BORROW_RECORD: Links members to books.
- FINE: Tracks late fees in RWF.
- WAITING_QUEUE: Members waiting for a book.
- AUDIT_LOG: Tracks all database changes.

PL/SQL Approach

- **Trigger:** trg_block_overdue stops a member with 3 overdue books from borrowing.
- **Function:** calculate_fine calculates (Days Late × 100 RWF).
- **Trigger:** trg_auto_fine creates a fine automatically when a book is returned late.
- **Procedure:** borrow_book saves the record and checks the rules.
- **Job:** DBMS_SCHEDULER runs every night to check for overdue books.

Expected Results

- A fully working Oracle database.
- No more lost book records.
- Automatic, fair fine calculation for all members.
- A live dashboard showing library activity.

Future Improvements and Limitations

- **Limitation:** The public dataset does not have Rwandan members, so I will create realistic sample data.

- **Future work:** Add a mobile app so students can check book availability from their phones.

References

- Kumar, A., & Singh, R. (2020). Design and Implementation of Library Management System.
- Ziegler, C. (2017). Book-Crossing Dataset. Kaggle.
- Oracle Corporation. (2024). Database PL/SQL Language Reference.